

Remediation Guide

Vulnerability remediation plan

Report Type	Remediation
Tool Source	Unified Multi-Tool
Reporting Month	July 2026

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1. Portfolio Risk Overview

Tool Source: Unified Multi-Tool

Reporting Month: July 2026

Total Open Findings: 160

Measure	Value
Consolidated Open	160
Affected Assets	85
Immediate Patch (P1 + P2)	113
Exploit Available	77
Cross-tool Confirmed	40
Single-source Only	120
Confirmation Rate	25%

Patch Priority Posture

Priority	Open Findings
P1	44
P2	69
P3	23
P4	24

Scanner Contribution

Scanner	Observed	P1 + P2	Exploit Available	Cross-tool Confirmed	Source-only
Tenable.sc	40	30	30	40	0
Tenable.io	40	30	30	40	0
Qualys	40	30	30	40	0
CrowdStrike	120	83	47	0	120

Highest-Risk Assets

Asset	Open	P1 + P2	Critical	Exploit Available	Sources	Exposure
cs-asset-001.corp.example	3	3	1	3	1	1000
cs-asset-043.corp.example	3	3	1	3	1	940
cs-asset-031.corp.example	3	3	1	3	1	925
cs-asset-022.corp.example	3	3	1	3	1	895

cs-asset-034.corp.example	3	3	1	3	1	895
cs-asset-028.corp.example	3	3	1	3	1	825
cs-asset-040.corp.example	3	3	1	3	1	825
cs-asset-025.corp.example	3	3	0	3	1	808

2. Trend Analysis

Period	Total Open	New	Patched	P1	P2	P3	P4	Confirmed	Single-source
April 2026	150	0	0	42	62	23	23	40	110
May 2026	160	30	20	44	66	24	26	40	120
June 2026	165	25	20	44	73	23	25	40	125
July 2026	160	20	25	44	69	23	24	40	120

3. Remediation Actions

1. Google Chrome use-after-free vulnerability

- Affected Findings: 15 across 15 assets
- Asset Examples: cs-asset-001.corp.example, cs-asset-003.corp.example, cs-asset-006.corp.example, cs-asset-011.corp.example, cs-asset-014.corp.example
- CVE: CVE-2024-4671
- Severity: Critical
- Patch Priority: P1
- Advisory: <https://chromereleases.googleblog.com/>
- Advisory: <https://nvd.nist.gov/vuln/detail/CVE-2024-4671>

Remediation steps:

1. Update Google Chrome to the latest enterprise stable release and relaunch the browser. Patch Google Chrome during the approved change window and capture validation evidence. Apply the minimum vendor-supported fixed release. Test in pre-production, deploy, restart if required, and run a verification scan.
2. Validate application and service health after the change.
3. Run a follow-up vulnerability scan and retain evidence.

```
winget upgrade --id Google.Chrome --exact --silent --accept-source-agreements --accept-package-agreements (Get-Item 'C:\Program Files\Google\Chrome\Application\chrome.exe').VersionInfo.FileVersion
```

Rollback and validation:

- Use the approved change rollback procedure if service-health checks fail.

- Confirm the installed version or configuration, validate service health, and verify closure in a follow-up scan.

2. SMB Signing Not Required

- Affected Findings: 4 across 4 assets
- Asset Examples: asset-007.corp.local, asset-017.corp.local, asset-027.corp.local, asset-037.corp.local
- CVE: N/A
- Severity: Critical
- Patch Priority: P1
- Advisory: <https://learn.microsoft.com/windows/security/threat-protection/security-policy-settings/microsoft-network-server-digitally-sign-communications-always>

Remediation steps:

1. Require SMB signing through Group Policy and validate domain controller and member server policy refresh.
2. Validate application and service health after the change.
3. Run a follow-up vulnerability scan and retain evidence.

```
# Replace placeholders with the approved product-specific command
<apply-approved-security-update>
<validate-service-health>
<run-follow-up-scan>
```

Rollback and validation:

- Use the approved change rollback procedure if service-health checks fail.
- Confirm the installed version or configuration, validate service health, and verify closure in a follow-up scan.

3. Apache Tomcat AJP Request Injection

- Affected Findings: 19 across 19 assets
- Asset Examples: asset-001.corp.local, asset-011.corp.local, asset-021.corp.local, asset-031.corp.local, cs-asset-002.corp.example
- CVE: CVE-2020-1938
- Severity: Critical
- Patch Priority: P1
- Advisory: <https://tomcat.apache.org/security-9.html>
- Advisory: <https://nvd.nist.gov/vuln/detail/CVE-2020-1938>

Remediation steps:

1. Apply the vendor security update, restart the affected service if required, and validate with a follow-up scan.
2. Validate application and service health after the change.
3. Run a follow-up vulnerability scan and retain evidence.

```
/opt/tomcat/bin/version.sh
grep -n 'Connector.*AJP' /opt/tomcat/conf/server.xml
# Upgrade Tomcat and disable unused AJP or require a secret
sudo systemctl restart tomcat
```

Rollback and validation:

- Use the approved change rollback procedure if service-health checks fail.
- Confirm the installed version or configuration, validate service health, and verify closure in a follow-up scan.

4. Remote Desktop Weak Encryption Enabled

- Affected Findings: 4 across 4 assets
- Asset Examples: asset-005.corp.local, asset-015.corp.local, asset-025.corp.local, asset-035.corp.local
- CVE: N/A
- Severity: Critical
- Patch Priority: P1
- Advisory: <https://www.tenable.com/plugins>
- Advisory: <https://www.qualys.com/research/security-alerts/>

Remediation steps:

1. Apply the vendor security update, restart the affected service if required, and validate with a follow-up scan.
2. Validate application and service health after the change.
3. Run a follow-up vulnerability scan and retain evidence.

```
Get-HotFix | Sort-Object InstalledOn -Descending | Select-Object -First 10
# Install the approved Microsoft security update and reboot if required
Get-Service TermServLicensing
```

Rollback and validation:

- Use the approved change rollback procedure if service-health checks fail.
- Confirm the installed version or configuration, validate service health, and verify closure in a follow-up scan.

5. Linux Kernel Security Update Available

- Affected Findings: 19 across 19 assets
- Asset Examples: asset-009.corp.local, asset-019.corp.local, asset-029.corp.local, asset-039.corp.local, cs-asset-003.corp.example
- CVE: CVE-2024-1086
- Severity: Critical
- Patch Priority: P1

- Advisory: <https://www.tenable.com/plugins>
- Advisory: <https://www.qualys.com/research/security-alerts/>
- Advisory: <https://nvd.nist.gov/vuln/detail/CVE-2024-1086>

Remediation steps:

1. Apply the vendor security update, restart the affected service if required, and validate with a follow-up scan.
2. Validate application and service health after the change.
3. Run a follow-up vulnerability scan and retain evidence.

```
uname -r
sudo apt-get update
sudo apt-get install --only-upgrade linux-image-generic
sudo reboot
```

Rollback and validation:

- Use the approved change rollback procedure if service-health checks fail.
- Confirm the installed version or configuration, validate service health, and verify closure in a follow-up scan.

6. Microsoft SQL Server Unsupported Version

- Affected Findings: 4 across 4 assets
- Asset Examples: asset-003.corp.local, asset-013.corp.local, asset-023.corp.local, asset-033.corp.local
- CVE: N/A
- Severity: Critical
- Patch Priority: P1
- Advisory: <https://learn.microsoft.com/sql/sql-server/end-of-support>

Remediation steps:

1. Upgrade the affected software to a vendor-supported release and validate with a follow-up scan.
2. Validate application and service health after the change.
3. Run a follow-up vulnerability scan and retain evidence.

```
# Replace placeholders with the approved product-specific command
<apply-approved-security-update>
<validate-service-health>
<run-follow-up-scan>
```

Rollback and validation:

- Use the approved change rollback procedure if service-health checks fail.
- Confirm the installed version or configuration, validate service health, and verify closure in a follow-up scan.

7. Windows Remote Desktop Licensing Service remote code execution

- Affected Findings: 15 across 15 assets
- Asset Examples: cs-asset-002.corp.example, cs-asset-005.corp.example, cs-asset-007.corp.example, cs-asset-010.corp.example, cs-asset-015.corp.example
- CVE: CVE-2024-38077
- Severity: Critical
- Patch Priority: P1
- Advisory: <https://msrc.microsoft.com/update-guide/vulnerability/CVE-2024-38077>
- Advisory: <https://nvd.nist.gov/vuln/detail/CVE-2024-38077>

Remediation steps:

1. Install the current Microsoft cumulative security update and restart the affected host. Patch Microsoft Windows Server during the approved change window and capture validation evidence. Apply the minimum vendor-supported fixed release. Test in pre-production, deploy, restart if required, and run a verification scan.
2. Validate application and service health after the change.
3. Run a follow-up vulnerability scan and retain evidence.

```
Get-HotFix | Sort-Object InstalledOn -Descending | Select-Object -First 10  
# Install the approved Microsoft security update and reboot if required  
Get-Service TermServLicensing
```

Rollback and validation:

- Use the approved change rollback procedure if service-health checks fail.
- Confirm the installed version or configuration, validate service health, and verify closure in a follow-up scan.

8. Cisco IOS XE web UI privilege escalation

- Affected Findings: 15 across 15 assets
- Asset Examples: cs-asset-002.corp.example, cs-asset-004.corp.example, cs-asset-007.corp.example, cs-asset-012.corp.example, cs-asset-015.corp.example
- CVE: CVE-2023-20198
- Severity: High
- Patch Priority: P1
- Advisory:
<https://sec.cloudapps.cisco.com/security/center/content/CiscoSecurityAdvisory/cisco-sa-iosxe-webui-privesc-j22SaA4z>
- Advisory: <https://nvd.nist.gov/vuln/detail/CVE-2023-20198>

Remediation steps:

1. Install the fixed Cisco IOS XE release and disable the HTTP server when it is not required. Patch Cisco IOS XE during the approved change window and capture validation evidence. Apply the minimum vendor-supported fixed release. Test in pre-production, deploy, restart if required, and run a verification scan.

2. Validate application and service health after the change.
3. Run a follow-up vulnerability scan and retain evidence.

```
show version
show running-config | include ^ip http
# Stage and activate the vendor-fixed IOS XE image under the approved network change
```

Rollback and validation:

- Use the approved change rollback procedure if service-health checks fail.
- Confirm the installed version or configuration, validate service health, and verify closure in a follow-up scan.

9. Oracle Java Unsupported Version

- Affected Findings: 4 across 4 assets
- Asset Examples: asset-008.corp.local, asset-018.corp.local, asset-028.corp.local, asset-038.corp.local
- CVE: CVE-2024-20919
- Severity: High
- Patch Priority: P1
- Advisory: <https://www.tenable.com/plugins>
- Advisory: <https://www.qualys.com/research/security-alerts/>

Remediation steps:

1. Upgrade the affected software to a vendor-supported release and validate with a follow-up scan.
2. Validate application and service health after the change.
3. Run a follow-up vulnerability scan and retain evidence.

```
# Replace placeholders with the approved product-specific command
<apply-approved-security-update>
<validate-service-health>
<run-follow-up-scan>
```

Rollback and validation:

- Use the approved change rollback procedure if service-health checks fail.
- Confirm the installed version or configuration, validate service health, and verify closure in a follow-up scan.

10. Apache Log4j Remote Code Execution

- Affected Findings: 19 across 19 assets
- Asset Examples: asset-002.corp.local, asset-012.corp.local, asset-022.corp.local, asset-032.corp.local, cs-asset-003.corp.example

- CVE: CVE-2021-44228
- Severity: High
- Patch Priority: P1
- Advisory: <https://logging.apache.org/log4j/2.x/security.html>
- Advisory: <https://nvd.nist.gov/vuln/detail/CVE-2021-44228>

Remediation steps:

1. Apply the vendor security update, restart the affected service if required, and validate with a follow-up scan.
2. Validate application and service health after the change.
3. Run a follow-up vulnerability scan and retain evidence.

```
find /opt -name 'log4j-core-*.jar' -print  
# Replace the vulnerable JAR through the approved application release  
sudo systemctl restart <affected-service>
```

Rollback and validation:

- Use the approved change rollback procedure if service-health checks fail.
- Confirm the installed version or configuration, validate service health, and verify closure in a follow-up scan.

11. Nginx Security Update Available

- Affected Findings: 4 across 4 assets
- Asset Examples: asset-006.corp.local, asset-016.corp.local, asset-026.corp.local, asset-036.corp.local
- CVE: CVE-2025-2345
- Severity: High
- Patch Priority: P1
- Advisory: https://nginx.org/en/security_advisories.html

Remediation steps:

1. Apply the vendor security update, restart the affected service if required, and validate with a follow-up scan.
2. Validate application and service health after the change.
3. Run a follow-up vulnerability scan and retain evidence.

```
# Replace placeholders with the approved product-specific command  
<apply-approved-security-update>  
<validate-service-health>  
<run-follow-up-scan>
```

Rollback and validation:

- Use the approved change rollback procedure if service-health checks fail.
- Confirm the installed version or configuration, validate service health, and verify closure in a follow-up scan.

12. TLS 1.0 Protocol Detection

- Affected Findings: 4 across 4 assets
- Asset Examples: asset-010.corp.local, asset-020.corp.local, asset-030.corp.local, asset-040.corp.local
- CVE: N/A
- Severity: High
- Patch Priority: P1
- Advisory: <https://www.tenable.com/plugins>
- Advisory: <https://www.qualys.com/research/security-alerts/>

Remediation steps:

1. Disable TLS 1.0 and TLS 1.1, enable TLS 1.2 or later, then restart the affected service.
2. Validate application and service health after the change.
3. Run a follow-up vulnerability scan and retain evidence.

```
# Replace placeholders with the approved product-specific command
<apply-approved-security-update>
<validate-service-health>
<run-follow-up-scan>
```

Rollback and validation:

- Use the approved change rollback procedure if service-health checks fail.
- Confirm the installed version or configuration, validate service health, and verify closure in a follow-up scan.

13. OpenSSL Security Update Available

- Affected Findings: 19 across 19 assets
- Asset Examples: asset-004.corp.local, asset-014.corp.local, asset-024.corp.local, asset-034.corp.local, cs-asset-001.corp.example
- CVE: CVE-2023-0286
- Severity: High
- Patch Priority: P1
- Advisory: <https://www.openssl.org/news/vulnerabilities.html>
- Advisory: <https://www.openssl.org/news/secadv/20230207.txt>
- Advisory: <https://nvd.nist.gov/vuln/detail/CVE-2023-0286>

Remediation steps:

1. Apply the vendor security update, restart the affected service if required, and validate with a follow-up scan.

2. Validate application and service health after the change.
3. Run a follow-up vulnerability scan and retain evidence.

```
openssl version -a
sudo apt-get update
sudo apt-get install --only-upgrade openssl
sudo systemctl restart <affected-service>
```

Rollback and validation:

- Use the approved change rollback procedure if service-health checks fail.
- Confirm the installed version or configuration, validate service health, and verify closure in a follow-up scan.

14. Microsoft Exchange privilege escalation

- Affected Findings: 15 across 15 assets
- Asset Examples: cs-asset-001.corp.example, cs-asset-004.corp.example, cs-asset-006.corp.example, cs-asset-009.corp.example, cs-asset-014.corp.example
- CVE: CVE-2024-21410
- Severity: Low
- Patch Priority: P2
- Advisory: <https://msrc.microsoft.com/update-guide/vulnerability/CVE-2024-21410>
- Advisory: <https://nvd.nist.gov/vuln/detail/CVE-2024-21410>

Remediation steps:

1. Install the supported Exchange Server security update and verify Extended Protection configuration. Patch Microsoft Exchange Server during the approved change window and capture validation evidence. Apply the minimum vendor-supported fixed release. Test in pre-production, deploy, restart if required, and run a verification scan.
2. Validate application and service health after the change.
3. Run a follow-up vulnerability scan and retain evidence.

```
Get-ExchangeServer | Format-List Name,Edition,AdminDisplayVersion
# Install the approved Exchange Security Update and reboot if required
```

Rollback and validation:

- Use the approved change rollback procedure if service-health checks fail.
- Confirm the installed version or configuration, validate service health, and verify closure in a follow-up scan.

4. Validation Requirements

Confirm service health, review change evidence, and verify closure in a follow-up vulnerability scan.

5. Reference Appendix

- <https://chromereleases.googleblog.com/>
- <https://nvd.nist.gov/vuln/detail/CVE-2024-4671>
- <https://learn.microsoft.com/windows/security/threat-protection/security-policy-settings/microsoft-network-server-digitally-sign-communications-always>
- <https://tomcat.apache.org/security-9.html>
- <https://nvd.nist.gov/vuln/detail/CVE-2020-1938>
- <https://www.tenable.com/plugins>
- <https://www.qualys.com/research/security-alerts/>
- <https://nvd.nist.gov/vuln/detail/CVE-2024-1086>
- <https://learn.microsoft.com/sql/sql-server/end-of-support>
- <https://msrc.microsoft.com/update-guide/vulnerability/CVE-2024-38077>
- <https://nvd.nist.gov/vuln/detail/CVE-2024-38077>
- <https://sec.cloudapps.cisco.com/security/center/content/CiscoSecurityAdvisory/cisco-sa-iosxe-webui-privesc-j22SaA4z>
- <https://nvd.nist.gov/vuln/detail/CVE-2023-20198>
- <https://logging.apache.org/log4j/2.x/security.html>
- <https://nvd.nist.gov/vuln/detail/CVE-2021-44228>
- https://nginx.org/en/security_advisories.html
- <https://www.openssl.org/news/vulnerabilities.html>
- <https://www.openssl.org/news/secadv/20230207.txt>
- <https://nvd.nist.gov/vuln/detail/CVE-2023-0286>
- <https://msrc.microsoft.com/update-guide/vulnerability/CVE-2024-21410>
- <https://nvd.nist.gov/vuln/detail/CVE-2024-21410>